# Consumer Outage Preparedness and Electricity Conservation

General educational guidance for consumer-facing SmartGrid AI advisories

**Document purpose:** This document is prepared as a searchable knowledge-base source for a SmartGrid AI RAG prototype. It is an educational synthesis, not an official ERCOT, utility, or government publication.

## Purpose

This document provides general educational information for consumer advisories. It is not a substitute for instructions from local emergency management, utilities, public health authorities, or emergency services. Recommendations should be adapted to local conditions and official guidance.

## Before an outage

Keep flashlights, batteries, a charged power bank, and a battery-powered or hand-crank radio available. Know how to report an outage to the electric utility and how to receive local emergency alerts.

People who depend on electrically powered medical equipment should discuss backup power and emergency plans with their healthcare provider and utility before an outage. Do not wait until an emergency to identify critical needs.

Keep refrigerators and freezers closed during an outage to preserve cold temperatures. Have a plan for safe drinking water, medications that require refrigeration, pets, and household members who may need assistance.

## During an outage

Stay away from downed power lines and anything touching them. Assume every downed line is energized and report it to the utility or emergency services. Never attempt to move a power line.

Use generators outdoors, well away from doors, windows, and vents, and follow the manufacturer's instructions. Never operate a generator, charcoal grill, or fuel-burning camping stove inside a home, garage, or enclosed space because carbon monoxide can be deadly.

Use battery-powered lighting rather than candles where possible. Unplug sensitive electronics if appropriate and follow utility instructions. Avoid opening refrigerators and freezers unnecessarily.

## Voluntary electricity conservation

When the grid operator or utility asks customers to conserve, customers may be able to reduce nonessential electricity use. Examples include adjusting thermostat settings modestly when safe, delaying use of high-energy appliances, turning off unnecessary lights, and shifting flexible activities away from peak periods.

Conservation advice should never compromise health or safety. Do not turn off medically necessary equipment, essential heating or cooling for vulnerable people, or safety systems. Follow official instructions if conditions become dangerous.

## How SmartGrid AI should phrase advice

Advice should be conditional, specific, and non-alarmist. For example: 'The model estimates elevated grid-stress risk for the next-day period. Consider delaying flexible high-energy activities if convenient, and follow your utility's official updates.'

Do not say 'A blackout will happen' based only on a model risk score. Do not imply that household conservation guarantees prevention of an outage. Include the forecast period and direct consumers to official utility and emergency-management information.

## Key takeaways

- Prepare supplies and an outage plan before an event occurs.
- Never approach downed power lines; treat them as energized.
- Use generators outdoors and prevent carbon-monoxide exposure.
- Conservation advice must not compromise health, safety, or essential needs.

## References and further reading


• Ready.gov Power Outages — https://www.ready.gov/power-outages

• U.S. CDC Carbon Monoxide Poisoning — https://www.cdc.gov/carbon-monoxide/

• ERCOT Demand Response — https://www.ercot.com/services/programs/load


*Prepared 9 November 2026. Verify current operational information with the linked official sources before making real-world decisions.*